



Traffic Engineering (TE) & Multiprotocol Label Switching (MPLS)

Redes de Comunicações II

Licenciatura em Engenharia de
Computadores e Informática

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Traffic supported by a Network

- A **traffic flow** from a sender host A to a receiver host B is the sequence of IP packets sent by host A to host B during time
 - The **throughput** is the rate, in bits/second, of the traffic flow
-
- Example: host A downloads a file of 5 GBytes from host B through TCP at an average download rate of 5000 IP packets/second.
 - Average download throughput of the file download:

Assuming that each IP download packet has 20 Bytes (IP header) + 20 Bytes (TCP header) + 1460 Bytes (Data) = 1500×8 bits, then, the download throughput is $1500 \times 8 \times 5000 = 60000000$ bps = 60 Mbps (Megabits/sec)
 - Average upload throughput of the file download:

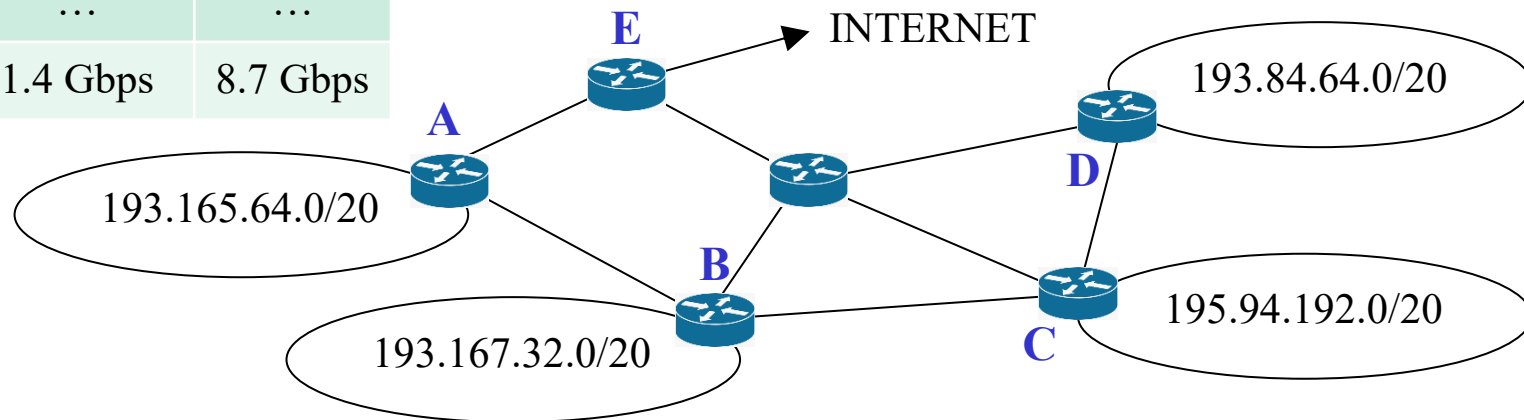
Assuming that each 2 downloaded TCP segments generate one upload TCP segment (to acknowledge the received data) in an IP packet with 20 Bytes (IP header) + 20 Bytes (TCP header) = 40×8 bits, then, the upload throughput is $40 \times 8 \times 2500 = 800000$ bps = 800 Kbps (Kilobits/sec)
 - File download time:

The file size 5 GBytes is segmented in $= 5 \times 1024^3 / 1460 \approx 3677$ TCP segments. So, the download time is around $3677 / 5000 = 0.74$ seconds

Traffic supported by a Network

- An **aggregated flow** from one network element A to another element B is the set of traffic flows from A to B.
 - The **throughput** of an aggregated flow is the total rate, in bits/second, of all its traffic flows
- The **traffic throughput of a service** supported by a network is defined (in its simplest version) by the average throughput over time of the aggregated flows between all node pairs running the service.

N1	N2	N1→N2	N2→N1
A	B	2.5 Gbps	3.1 Gbps
A	C	1.3 Gbps	1.7 Gbps
...	...	...	...
D	E	1.4 Gbps	8.7 Gbps



Network Dimensioning vs. Traffic Engineering

- **Network Dimensioning** is the process of determining the required capacity on each element of the network to be able to carry the traffic throughput of all supported services
 - Network elements are links, routers, switches, etc...
 - Traffic throughputs are hard to predict!
- **Traffic Engineering** is the process selecting the routing paths to best fit the traffic of all supported services in the capacity resources of the network
 - Ensure the most desirable/appropriate routing paths for each aggregated flow of each service.
 - The aim is to use efficiently the capacity resources of the network.
 - For example, to prevent over-utilized (congested) links whilst other links are under-utilized.

Life Cycle of a Communications Network

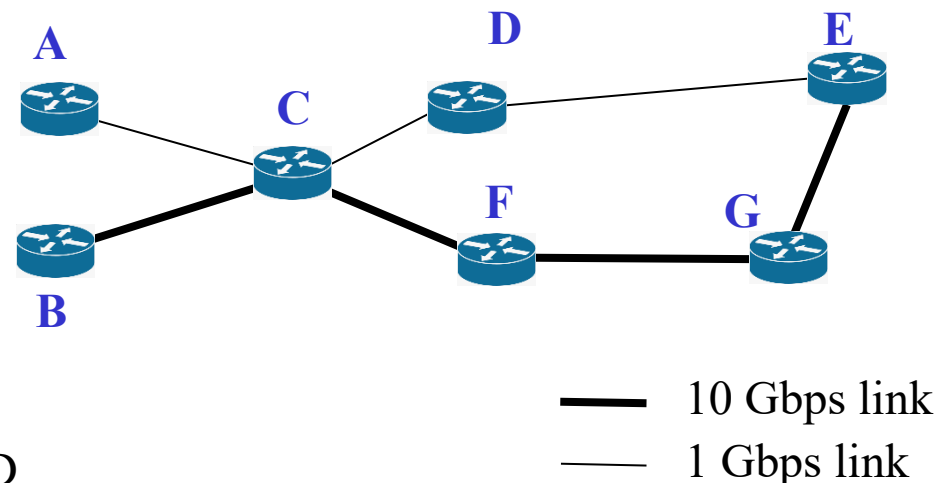
- A network life cycle is composed by two management phases:
 - A network dimensioning phase
 - The operator estimates the traffic throughput of the services the network must support in a given future time (for example, in the next year).
 - Then, it determines the required capacity of the network elements to carry the estimated traffic throughputs.
 - The aim is to minimize the investment cost.
 - Finally, the operator upgrades the network.
 - A traffic engineering phase
 - The operator changes the routing of the network in accordance with the traffic changes of the supported services.
 - This phase continues while network resources are enough to support all services
 - An efficient traffic engineering allows this phase to last for a longer time (minimizing the need for a new investment cost).
 - A new network dimensioning phase is required when the network capacity resources are no longer enough to support all services due to:
 - (i) throughput grow of the existing aggregated flows, (ii) new aggregated flows of the initial services and/or (iii) the introduction of new services.

Traffic Engineering with IGP Routing Protocols

- Traffic engineering is not ideal when using standard IGP protocols (as RIP or OSPF)

- With RIP, all flows are routed in C through next-hop D
 - There is a huge overflow (i.e., a drop) of 10 Gbps in link C-D
- With OSPF, and assigning appropriate OSPF port costs:
 - All flows can be routed in C through next-hop F
 - overflow of 1 Gbps in link C-F
 - All flows can be routed in C through next-hops D and C (with ECMP)
 - overflow of 4.5 Gbps in link C-D

N1	N2	N1→N2	N2→N1
A	E	1.0 Gbps	-
B	E	5.0 Gbps	-
C	E	5.0 Gbps	-



Traffic Engineering with IGP Routing Protocols

- In order to fit the traffic flow throughputs in the capacity resources of the network, we need to route each traffic flow through:

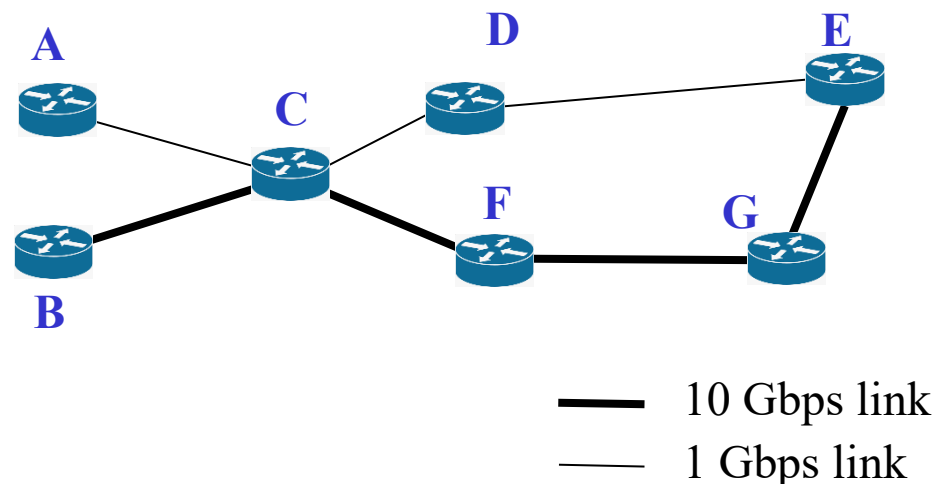
A – E: $A \rightarrow C \rightarrow D \rightarrow E$

B – E: $B \rightarrow C \rightarrow F \rightarrow G \rightarrow E$

C – E: $C \rightarrow F \rightarrow G \rightarrow E$

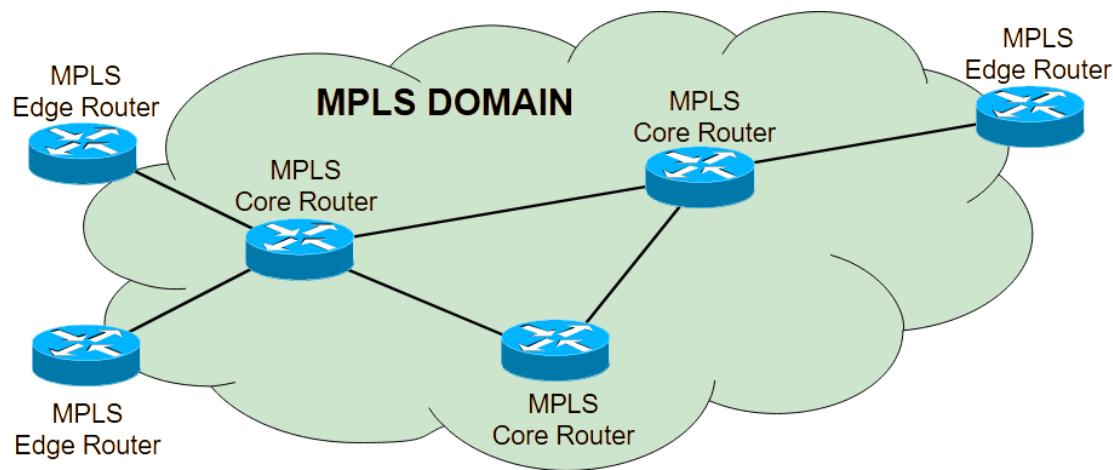
N1	N2	N1→N2	N2→N1
A	E	1.0 Gbps	-
B	E	5.0 Gbps	-
C	E	5.0 Gbps	-

- Note that:
 - this solution requires C to route flows based not only on the destination but also on the source
 - IGP protocols (as RIP or OSPF) are destination-based routing protocols



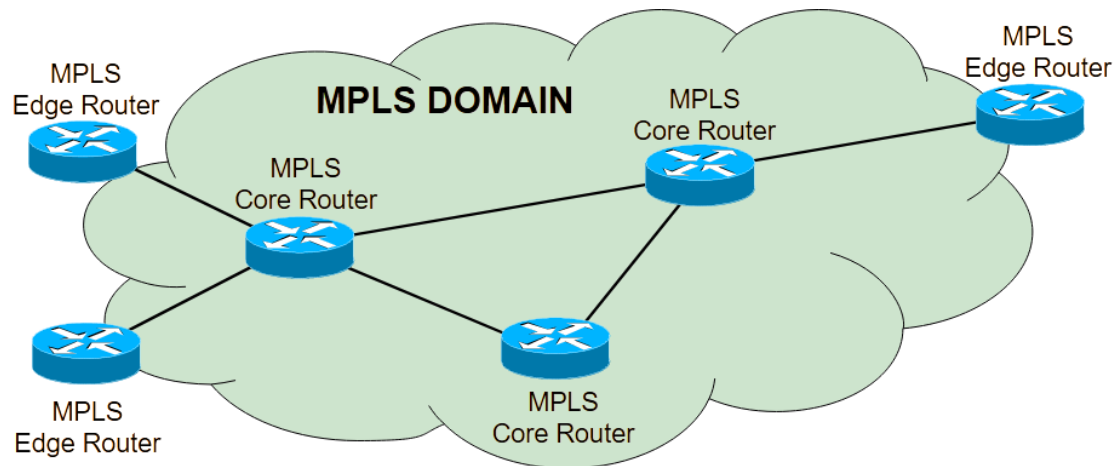
Multiprotocol Label Switching (MPLS)

- MPLS is a routing technique that forwards packets from one router to the next router based on labels rather than on network addresses.
 - While network addresses identify endpoints, labels identify established routing paths between endpoints.
 - MPLS can encapsulate packets of different protocols (IPv4 and IPv6), hence the ‘multiprotocol’ component of the name.
- In an **MPLS domain**, there are **Edge Routers** and **Core Routers**.

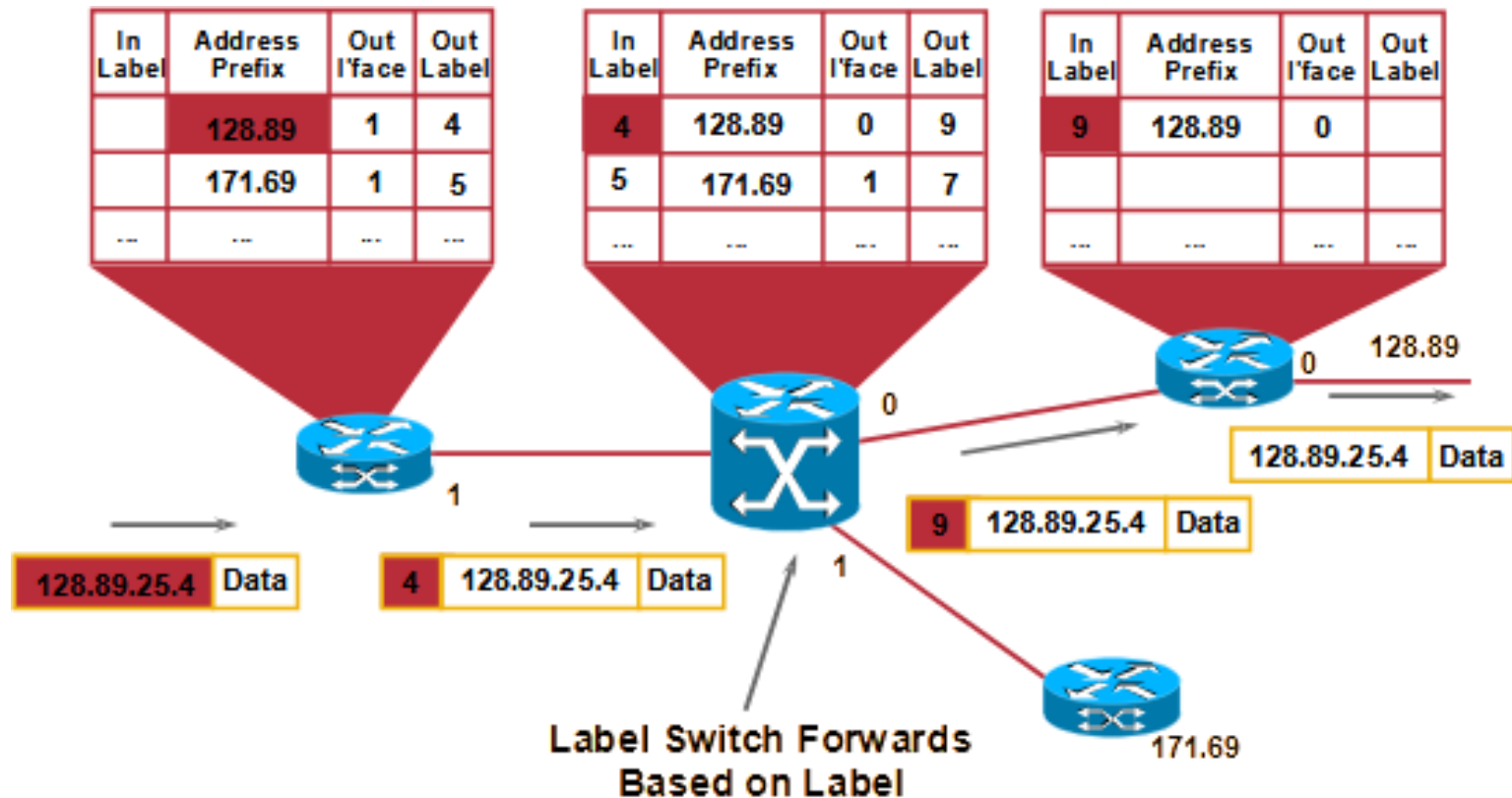


Multiprotocol Label Switching (MPLS)

- Packets are labelled at the entry Edge Router with a label of the first hop.
- Each Core Router forwards the packet based on its incoming label and swaps the label to the one of the next hop.
- The exit Edge Router removes the label.
- Advantages:
 - Simplification of the packet routing process on routers.
 - Full traffic engineering capability.
 - Simplification of the network management (a single protocol layer).

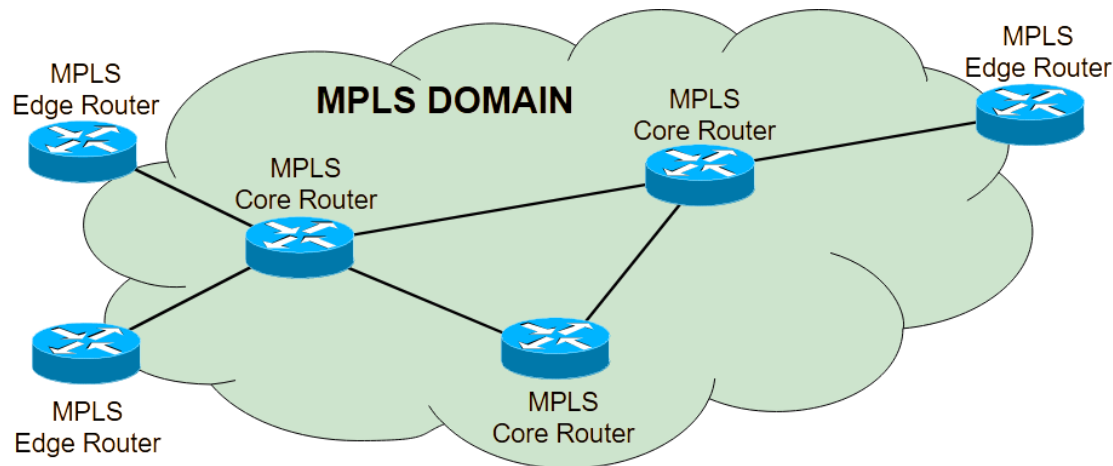


MPLS Switching Illustration



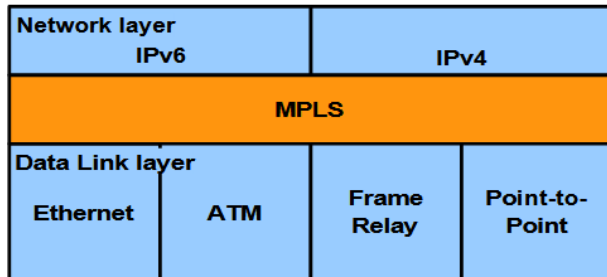
MPLS Fundamentals

- As a packet enters an MPLS domain, it is assigned a label based on its **Forwarding Equivalence Class (FEC)** defined at the entry **Edge Router**.
- FECs are groups of packet flows forwarded through the same **Label Switched Path (LSP)** by **Label Switching Routers (LSR)**.
 - A mechanism is needed to assign and distribute labels to establish LSP paths.
- In **Core Routers**, routing is based on label-swapping and forwarding.
- MPLS works in two planes:
 - Control Plane - responsible for setup label swapping tables among LSRs.
 - Forwarding Plane - uses the label carried by each incoming packet and the label swapping table of the LSR to forward the packet.

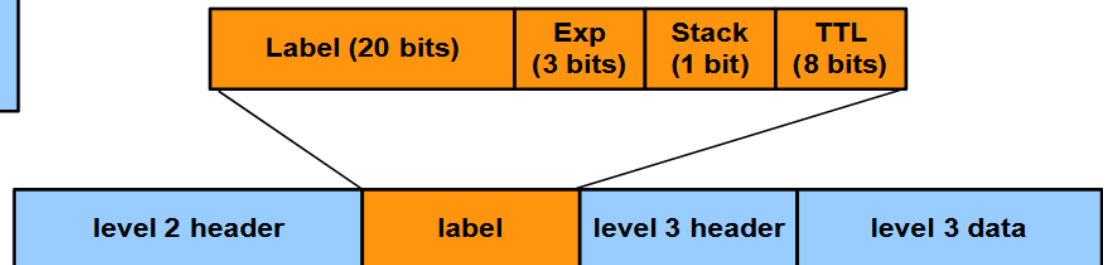


MPLS Labels

- Label is a 20-bit field that carries the value of the Label.
 - TTL field is IP independent – Similar purpose.
- On some Data Link (Layer 2) old technologies, label is given by the appropriate fields of their header.
 - ATM : VPI (Virtual Path ID) and VCI (Virtual Channel ID) fields.
 - Frame Relay: DLCI (Data Link Connection Identifier) field.
- On other Data Link technologies (Point-to-Point, Ethernet), the label is inserted between Layer 2 and Layer 3 headers.

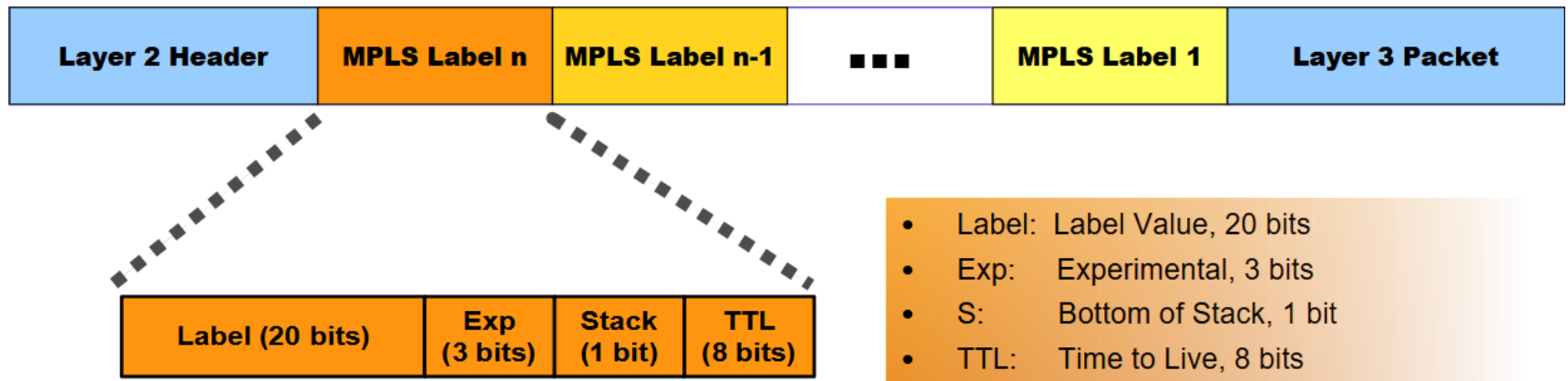


- Label: Label Value, 20 bits
- Exp: Experimental, 3 bits
- S: Bottom of Stack, 1 bit
- TTL: Time to Live, 8 bits



MPLS Label Stacking

- Labels can be used in stacks to support more efficiently multiple services:
 - Outer (last) label (first received) is used to switch the packets between Edge Routers in the MPLS core.
 - Inner labels are used to designate services, FECs, etc.
- S (Bottom of Stack) bit is set to one for the last entry in the label stack (i.e., for the bottom of the stack), and zero for all other labels.

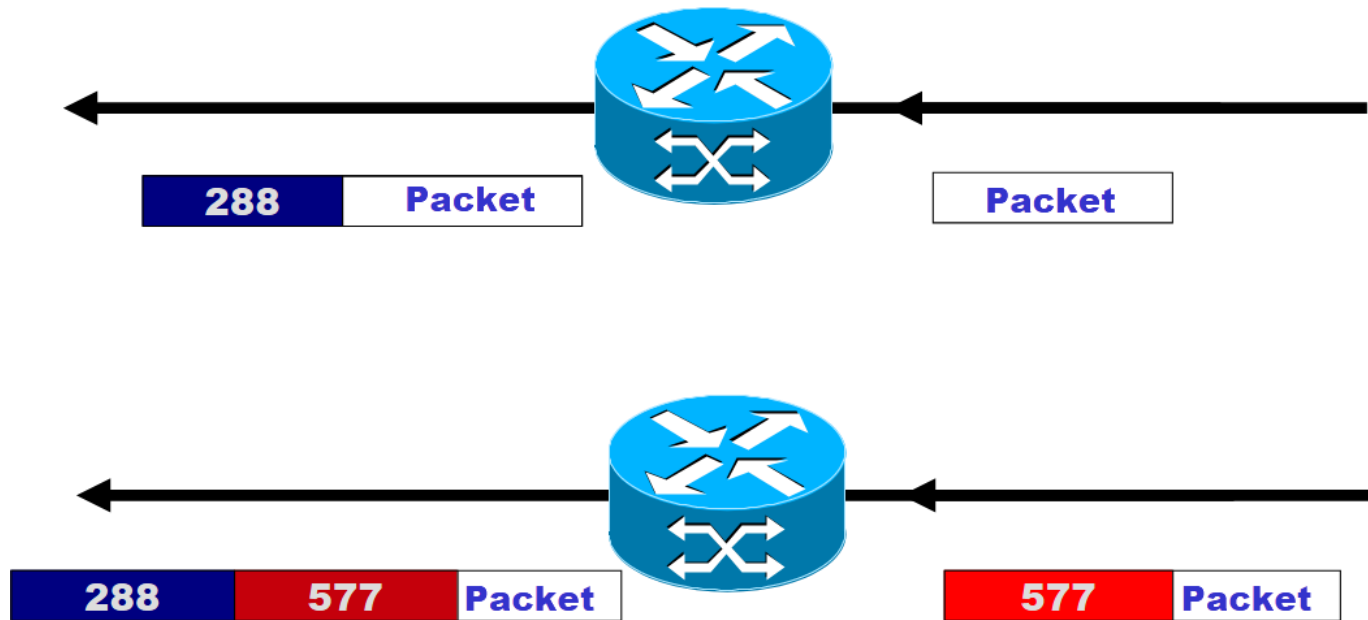


MPLS Label Swapping



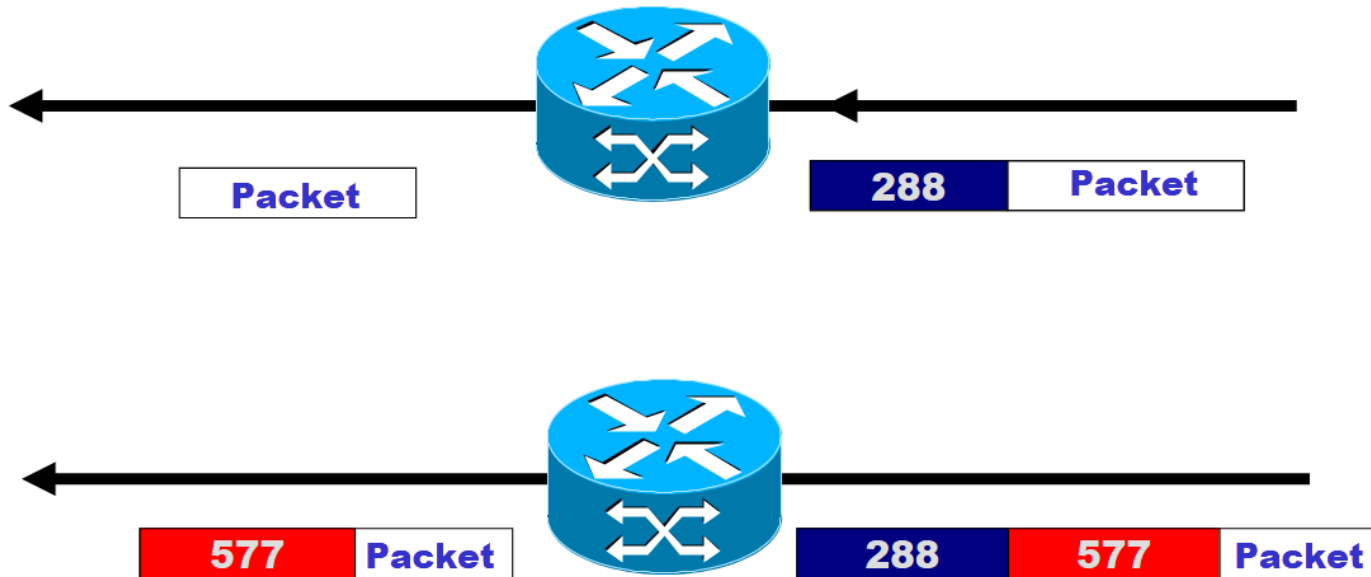
- Swapping is the process of changing the label of an incoming packet in the outgoing packet.
 - Labels are short and based on fixed-length fields.
 - Forwarding based on labels is much more efficient than forwarding based on IP routing tables.
 - Throughput capacity of MPLS routers is much higher than throughput capacity of IP routers.

MPLS Pushing Labels



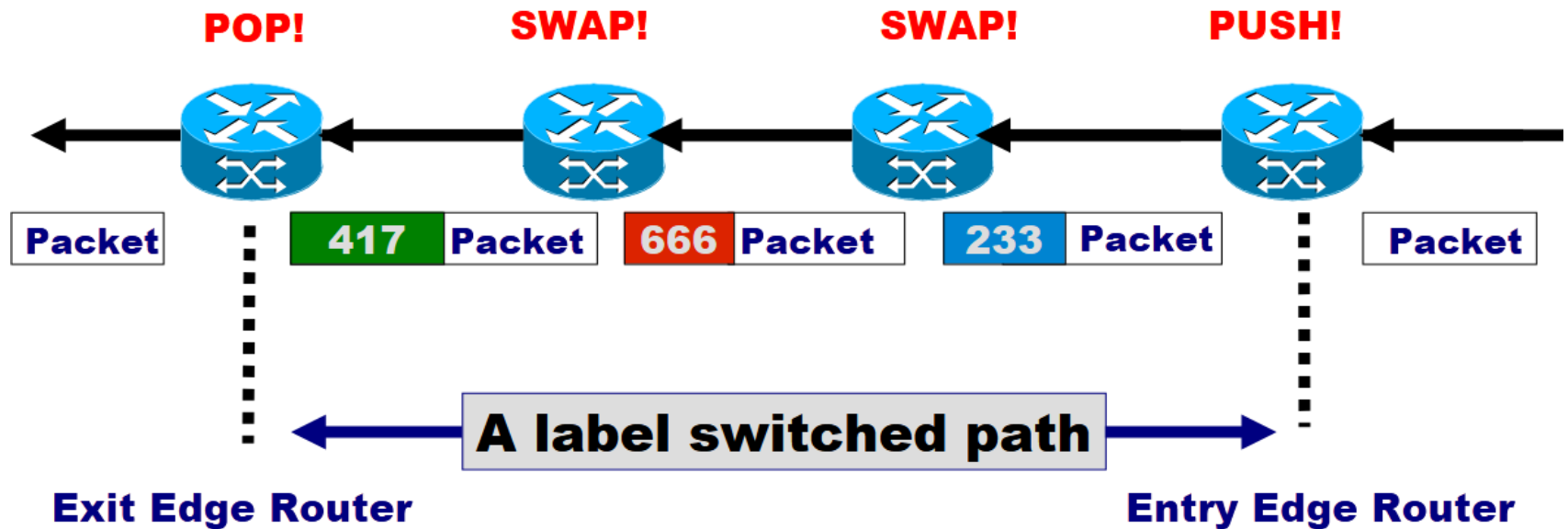
- Pushing is the process of adding a label to an incoming packet in the outgoing interface.

MPLS Popping Labels



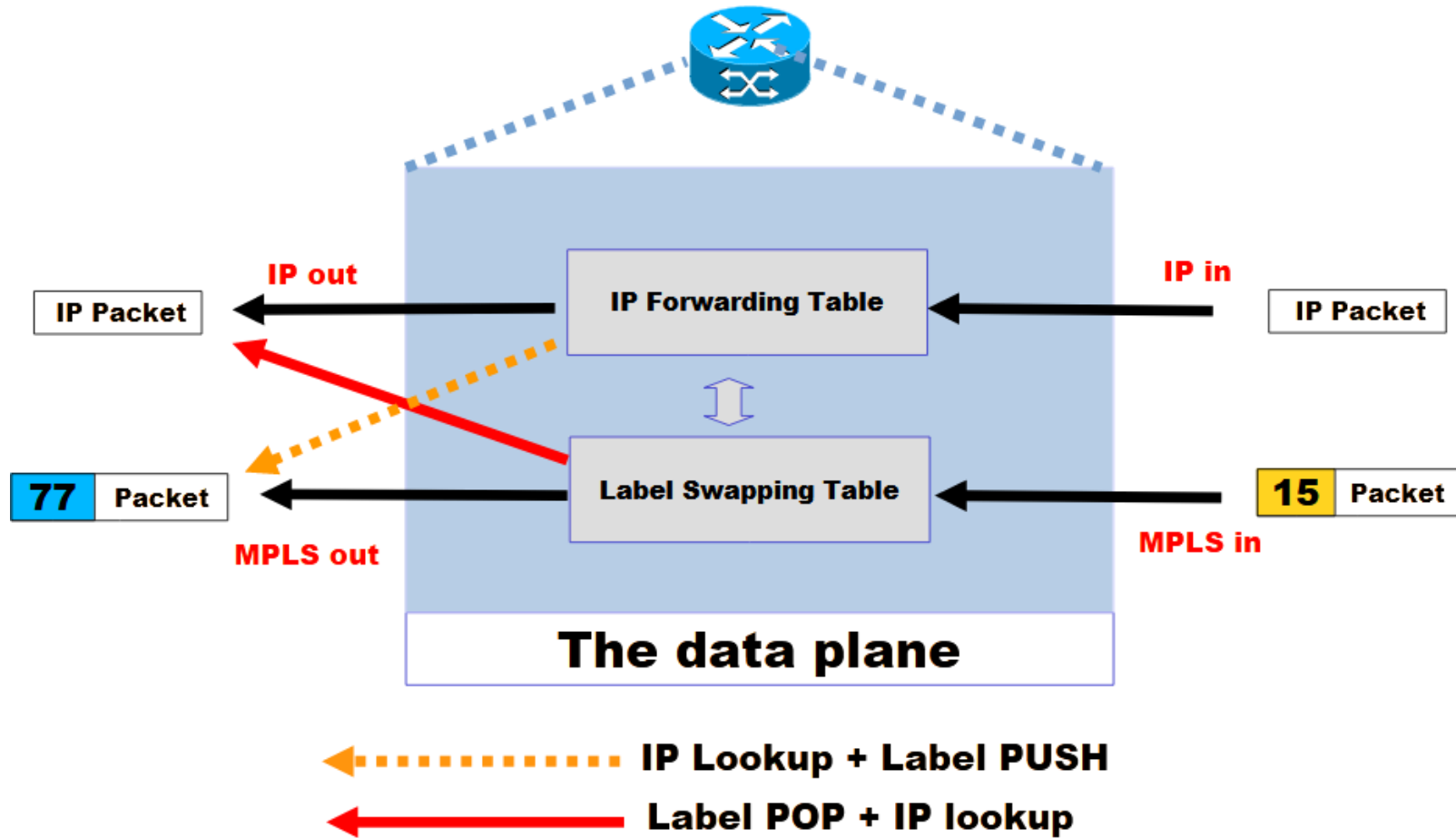
- Popping is the process of removing a label from an incoming packet in the outgoing interface.

A Label Switched Path (LSP)

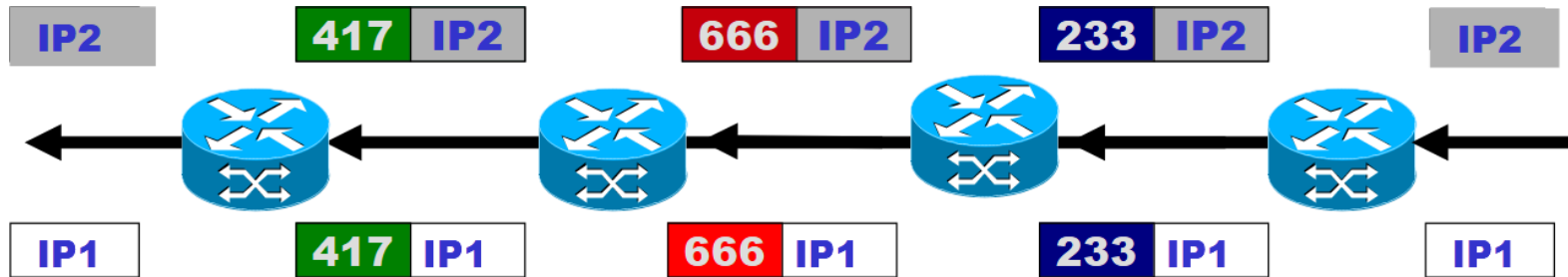


- An LSP is often referred to as a MPLS tunnel, as payload headers are not processed in the intermediate routers of an LSP

Label Switched Router (LSR)



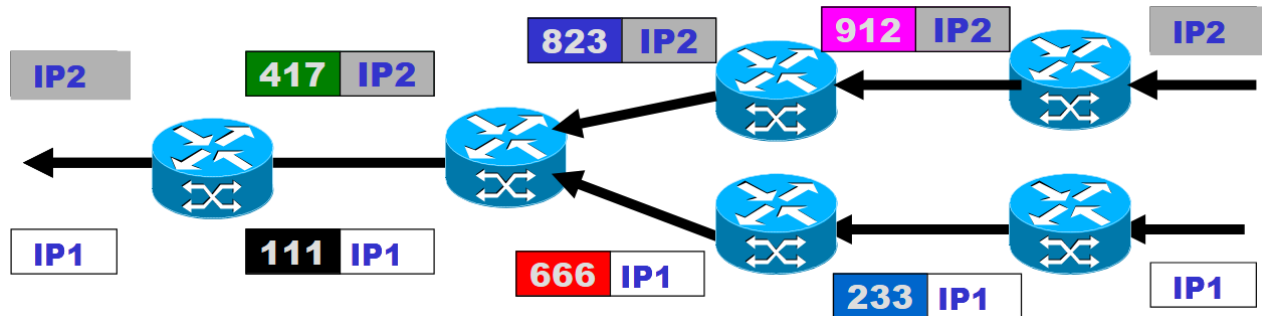
Forwarding Equivalence Class (FEC)



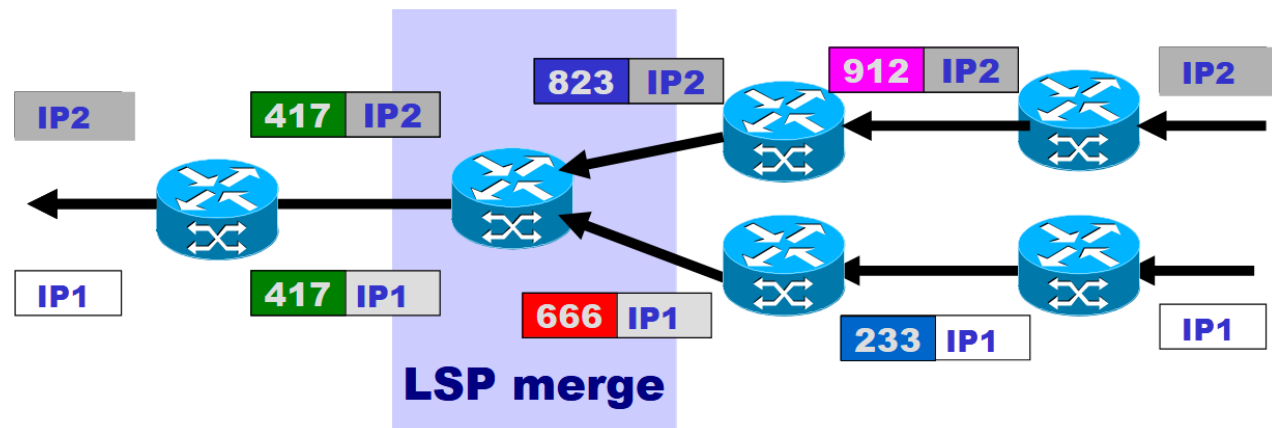
- Packets IP1 and IP2 are in the same FEC: they are forwarded in the same way.
 - IP layer headers are not processed inside an MPLS LSP.
 - This means that inside of the tunnel, the LSRs do not use IP routing tables for forwarding decisions.

Merging LSPs

- Two FECs to the same exit Edge Router with a common part in the two routing paths:

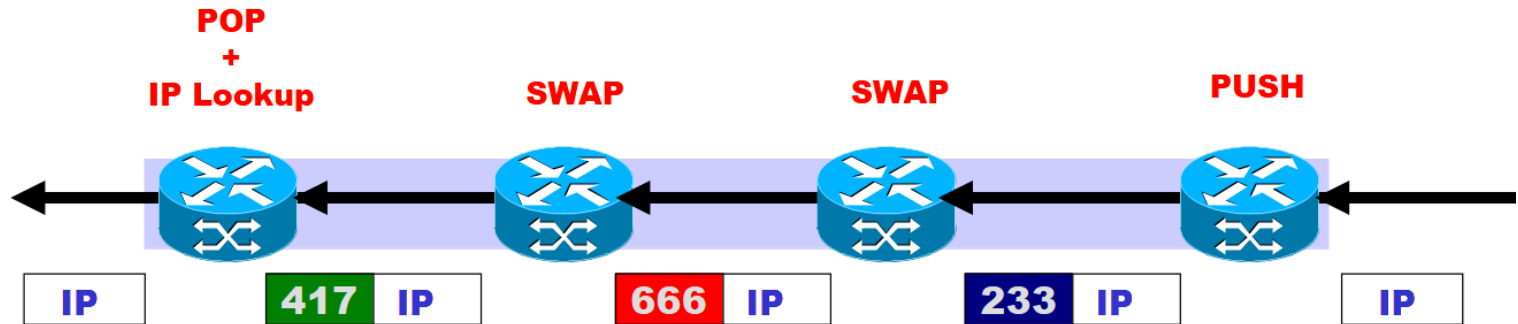


- The common part can be merged to reduce the size of the Label Swapping table:

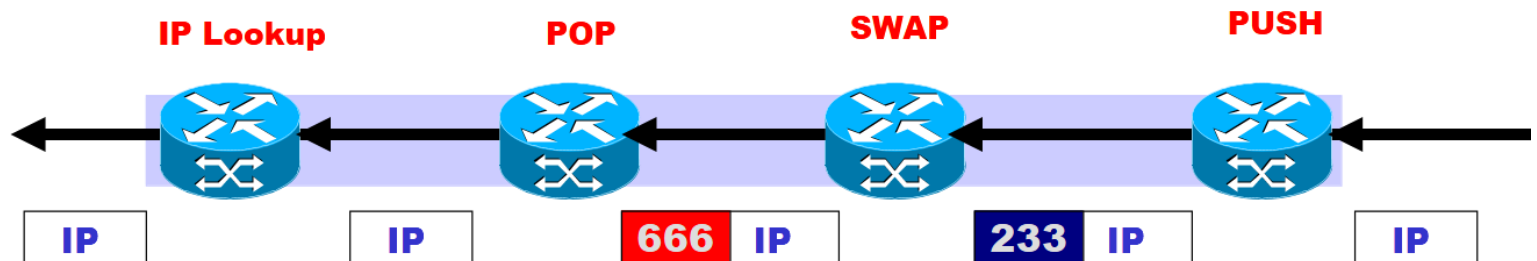


Penultimate Hop Popping (PHP)

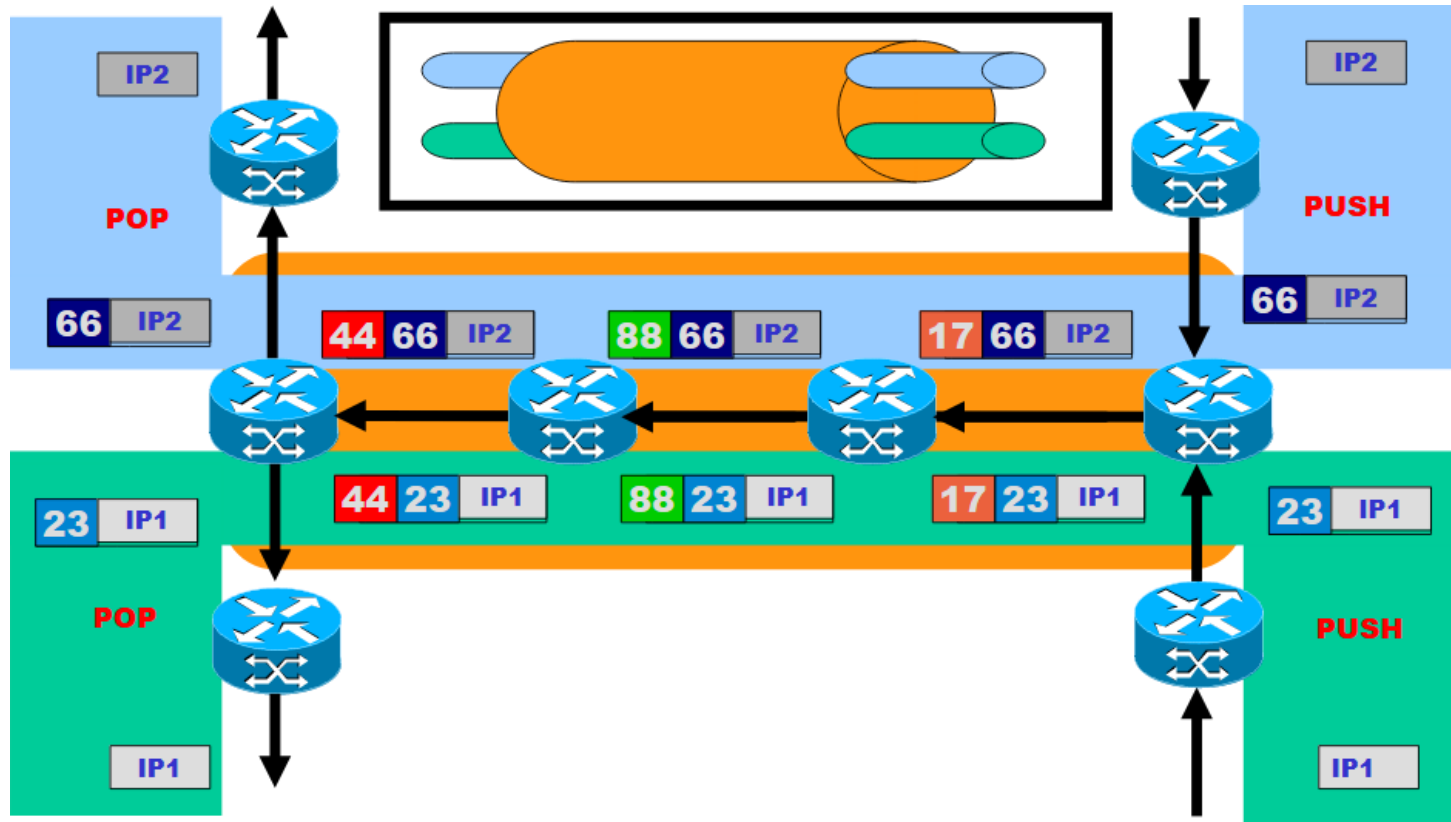
- An LSP from an entry Edge Router to an exit Edge Router:



- PHP is the process of label popping in the last Core Router before the packet reaches its exit Edge Router
 - it reduces the Edge Router load (usually, Edge Routers have more processing tasks than Core Routers)



LSP Hierarchy via Label Stacking



- The blue and the green LSPs are supported over the orange LSP in the central part of the MPLS network

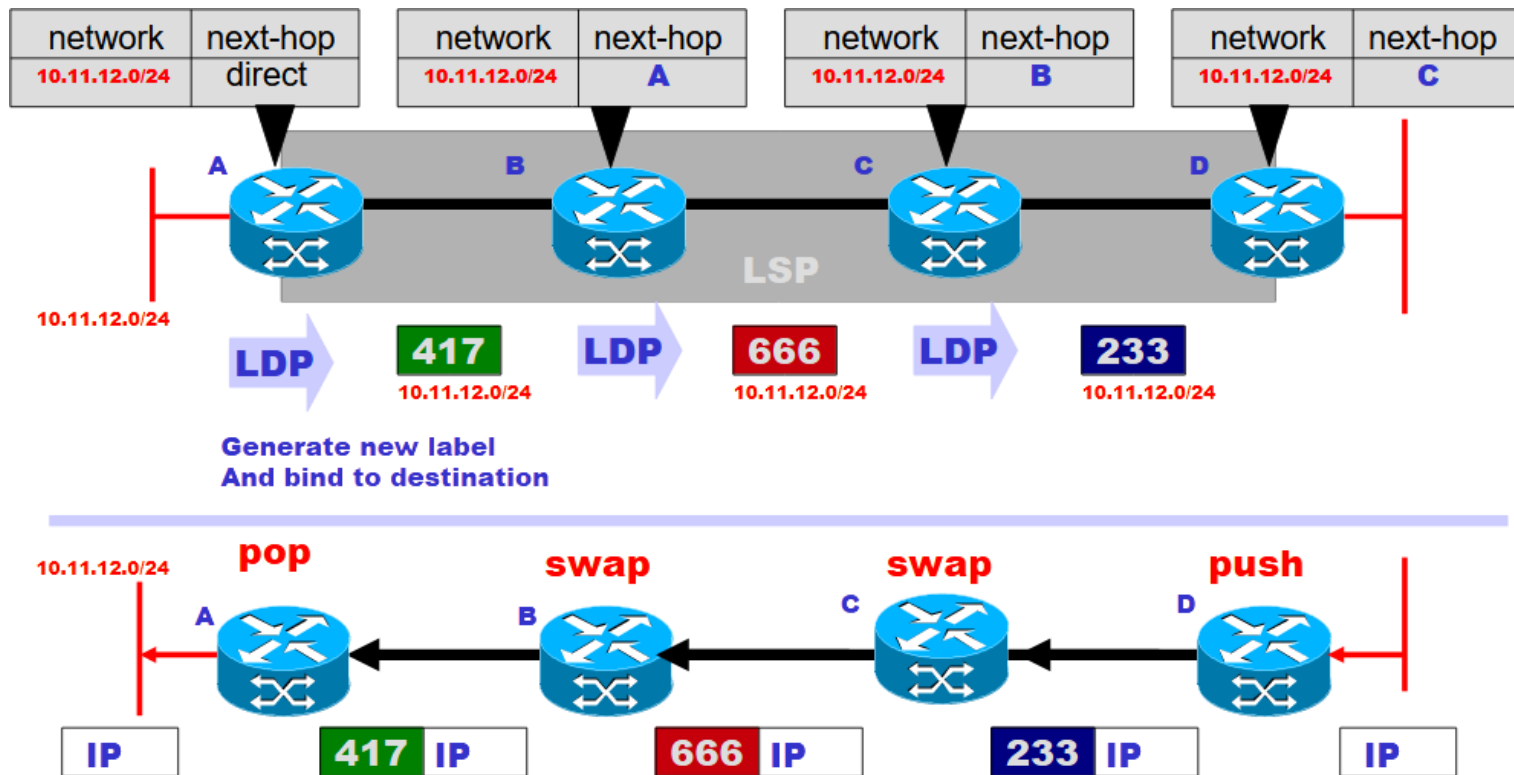
MPLS Control Plane: establishment of LSPs

- LSPs for best-effort services
 - All LSRs run a IGP protocol (OSPF, IS-IS).
 - Paths of LSPs are chosen based on shortest paths of IGP.
 - There is no traffic engineering (MPLS is used to make packet forwarding more efficient).
 - Label Distribution Protocol (LDP).
- LSPs with resource reservation
 - Constrained by explicit path definition and/or throughput availability on adopted paths.
 - Resource Reservation Protocol with Traffic Engineering (RSVP-TE).
 - Extension of RSVP to support traffic engineering and label distribution.
 - Constrained based Routing LDP (CR-LDP).
 - Extension of LDP to support constrained routing.
 - Deprecated!

Label Distribution Protocol (LDP)

- LDP specification is defined in RFC 5036 (Oct 2007)
 - There is a LDPv6 version to support IPv6, defined in RFC 7552: Updates to LDP for IPv6 (June 2015)
- LDP uses an IGP protocol running in LSRs for deciding the LSP routing paths
- LDP includes automatic discovery of LSRs (based on Hello messages running over UDP)
- Label mappings are exchanged over TCP for reliability.
- Label mappings are assigned by LSRs in the opposite direction of the LSPs (illustrated in the next slide)
- LDP considers different modes on how label mappings are assigned by LSRs to LSPs (so-called modes of behaviour)
 - Modes of behaviour are negotiated during session initialization

LDP and Hop-by-Hop routing



- In this example, an LSP is established from the ingress router **D** to the egress router **A** for the FEC defined as all IP packets destined to 10.11.12.0/24
- The mapping of each label to this LSP is assigned in the opposite direction of the LSP.

Modes of Behaviour of LDP

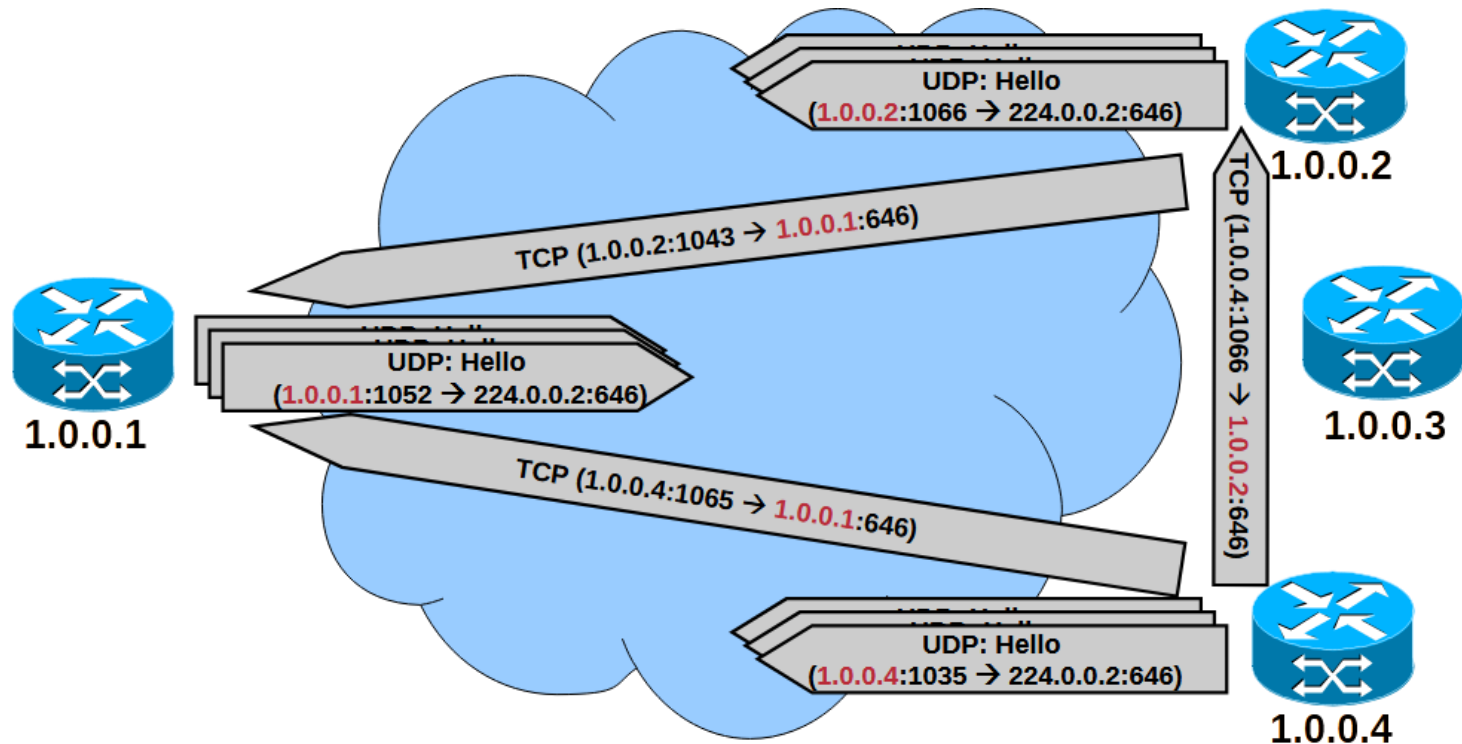
- Label distribution (independent or ordered)
 - Independent: each LSR may advertise label mappings to its neighbours at any time
 - Ordered: each LSR may advertise a label mapping only for a FEC for which it has a label mapping for the FEC next hop, or for which the LSR is the egress LSR
- Label advertisement (downstream unsolicited or on-demand)
 - Downstream Unsolicited: label mapping advertisements for all routes may be received from all neighbour LSRs
 - Downstream on Demand: LSRs request label mappings from next-hop LSR (according to IGP routing) and label mapping advertisements are sent upon requests
- Label retention (liberal or conservative)
 - Liberal: label mappings received from a neighbour LSR are retained regardless of whether the LSR is the next hop for the advertised mapping
 - Conservative: received label mappings are retained only if they are from a valid next-hop LSR (according to IGP routing)

LDP Messages

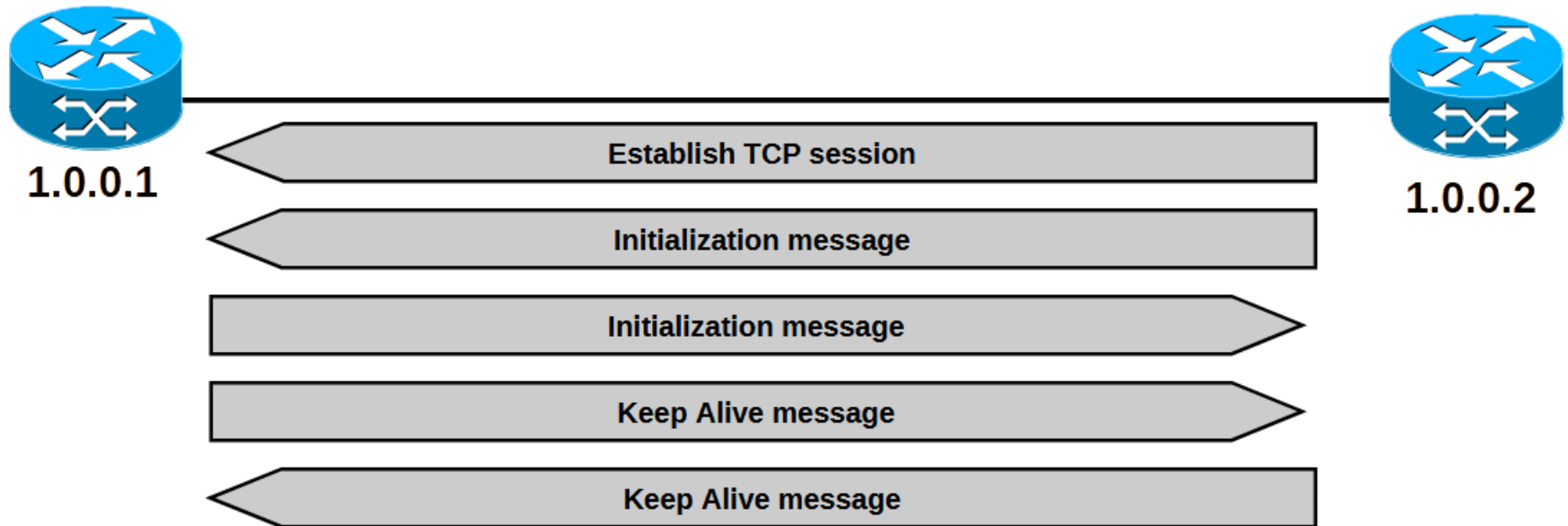
- Discovery messages
 - Announce and maintain the presence of an LSR in a network.
 - **Hello Messages** (UDP) are sent to “all-routers” multicast address (224.0.0.2).
 - Once neighbour is discovered, a LDP session is established over TCP.
- Session messages
 - Establish (**Initialization Message**) and maintain (**Keep Alive Message**) sessions between LDP peers.
- Advertisement messages
 - When a new LDP session is initialized and before sending label information, an LSR advertises its interface addresses with one or more **Address Messages**.
 - An LSR withdraws previously advertised interface addresses with **Address Withdraw Messages**.
 - Create, change, and delete label mappings for FECs, with **Label Mapping**, **Label Request**, **Label Abort Request**, **Label Withdraw**, and **Label Release Messages**.
- Notification messages
 - Provide advisory information and/or error information.

LDP Neighbour Discovery

- Hello messages (UDP) are periodically sent on all interfaces enabled for MPLS to a “all-routers” multicast address (224.0.0.2).
- On each network, when a pair of MPLS enabled routers know each other (each one has received a Hello message from the other), the router with the highest IP address establishes of an LDP/TCP session between them.
- Both TCP and UDP messages use well-known LDP port number 646.



LDP Session Negotiation



- Peers first exchange Initialization messages.
- The session is ready to exchange label mappings after receiving the first Keep Alive message.
 - Keep Alive messages are resent periodically to maintain the LDP/TCP session active.

MPLS Traffic Engineering

- Traffic Engineering is the process of selecting the paths of the LSPs that best fit the traffic in the capacity resources of the network
 - The previous LDP is appropriate to set up LSPs supporting best-effort services (TCP based services that work properly whatever network available resources)
 - LSPs with required reservation of resource (i.e., supporting services with required Quality of Service) need to be established through paths whose links still have enough available resources
- Selection of the paths of LSPs with resource reservation:
 - Offline: by a centralized management system that computes the adequate routing path for each LSP such that all reservations can be accepted
 - Online: through a Constrained-based Routing protocol that allows each ingress router of an LSP to compute a minimum cost path towards the LSP egress router whose links have enough available resources
- Establishment of an LSP with resource reservation over a given path:
 - Resource Reservation Protocol with Traffic Engineering (RSVP-TE)
 - Evolution of RSVP to support traffic engineering and label distribution
 - Constrained based Routing LDP (CR-LDP): deprecated

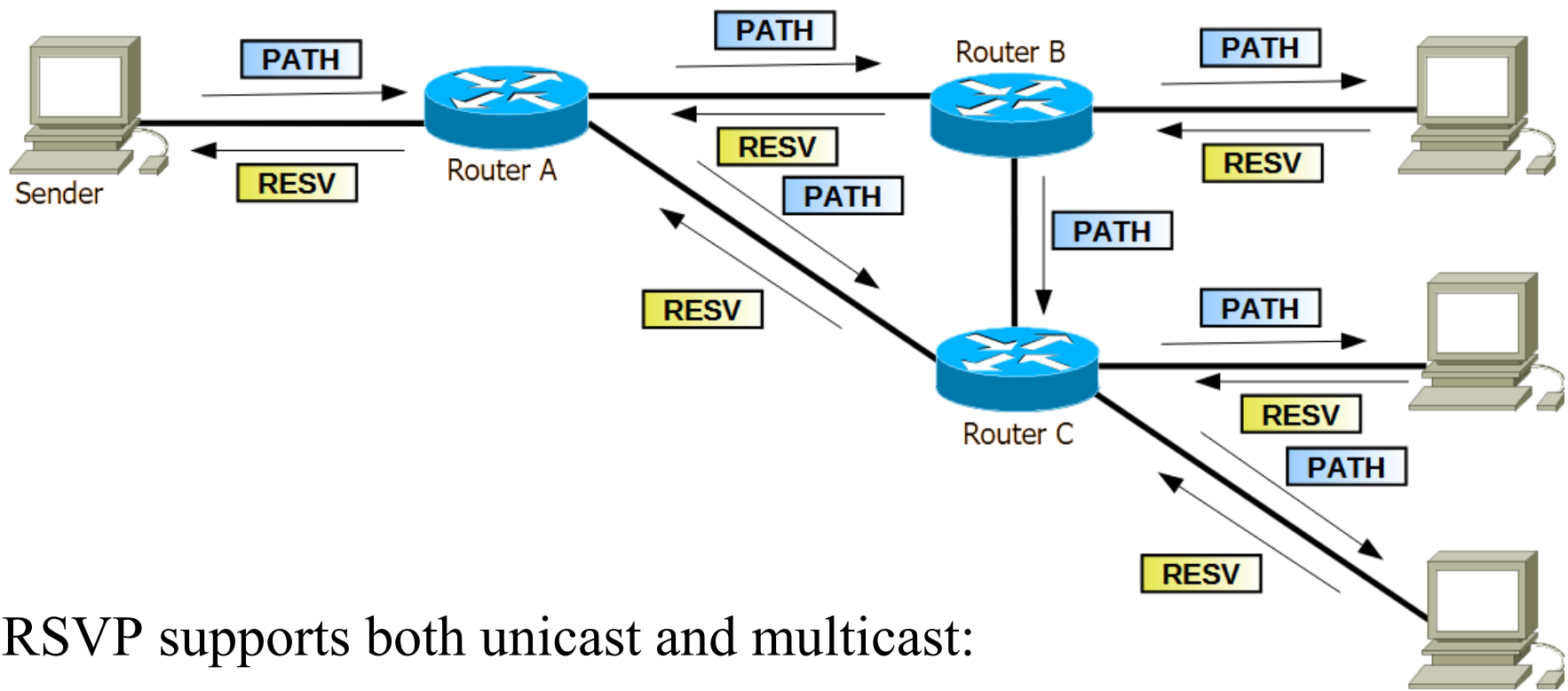
Resource Reservation Protocol (RSVP)

- RSVP is a signalling protocol that allows the reservation of network resources from source to destination host(s) of IP flows:
 - The source of each flow describes the characteristics of the IP packets.
 - The destination(s) of each flow describe the resources to be reserved in the network.
 - The network routers set up the requested reservation on their outgoing interfaces of the routing path from source to destination(s).
- Signalling is based on PATH and RESV messages:
 - PATH specifies the IP flow characteristics of the source.
 - RESV specifies the reservation requests of the destination(s).
 - If the reservation is not possible in a router, a RESV ERR message is sent by the router towards the destination(s).
- RSVP runs over IP (IP protocol type = 46).
- PATH and RESV messages are sent periodically to maintain the reservation state in the routers (soft states).

RSVP Messages

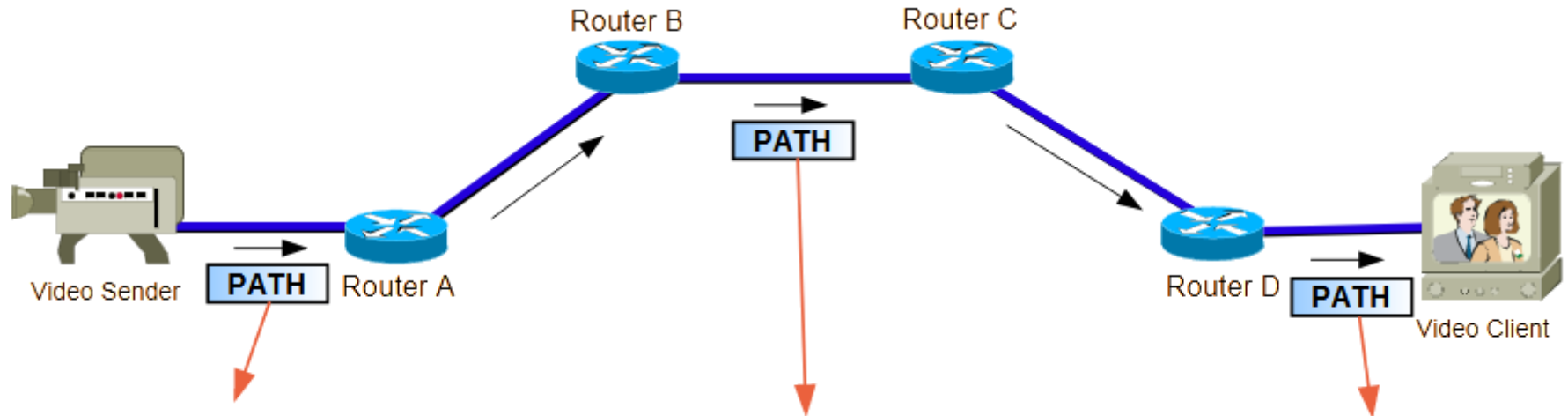
- PATH (Type = 0x01)
 - **Tspec** (“flow traffic specification”): contains the parameters that describe the traffic source based on the “Token Bucket” model
- RESV (Type = 0x02)
 - **Tspec**: the same that was received on the PATH message
 - **FilterSpec** (“filter specification”): contains the flow descriptor that enables routers to identify the IP packets that belong to the reservation (source address, destination address, protocol type, source port number, destination port number, any combination of these parameters)
 - **Rspec** (“flow reservation specification”): contains the parameters describing the reservation requested by the destination
 - Rspec is specified if the destination wants a service of type “guaranteed service” : it specifies the bandwidth resources to be reserved in the path
 - When Rspec is not specified, it means that the destination wants a service of type “controlled load” : it is up to the routers to decide the bandwidth resource to be reserved so that the IP packet flow does not suffer congestion.

RSVP signalling



- RSVP supports both unicast and multicast:
 - Both messages are routed based on the IP routing protocol
 - Multicast routing protocols will be the next module of this course unit
 - PATH messages: paths are selected by the routers
 - RESV messages: routers set up the reservation on the selected paths
 - In multicast, routers make reservation aggregation

RSVP PATH (Example)

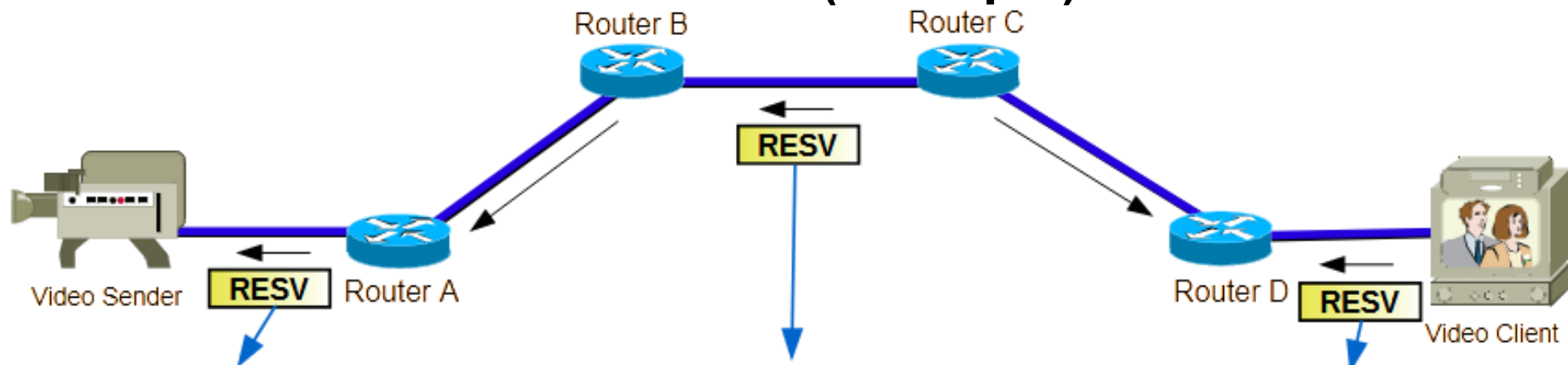


Vs.: 4	iHL: 5	Service	Total Length: 60	
Identification			Flg	Fragment Offset
Time to Live		Protocol: 46	Header Checksum	
Source Address:			Video Server	
Destination Address:			Video Client	
1	0	Type: 1	Checksum	
Send_TTL		0	Message Length: 40	
SESSION Length.: 12			Class Nº: 1	Class Type: 1
Destination Address:			Video Client	
Protocol ID		Flags	Destination port	
RSVP_HOP Length. : 12			Class Nº: 3	Class Type: 1
Last Hop Address:			Video Server	
Logical Interface Handle of the last node (LIH)				
TIME_VALUES Length: 8			Class Nº: 5	Class Type: 1
Update Period (ms)				

Vs.: 4		iHL: 5		Service		Total Length: 60			
Identification						Flg	Fragment Offset		
Time to Live			Protocol: 46			Header Checksum			
Source Address:						Video Server			
Destination Address:						Video Client			
1	0	Type: 1			Checksum				
Send_TTL			0			Message Length: 40			
SESSION Length: 12						Class Nº: 1		Class Type: 1	
Destination Address:						Video Client			
Protocol ID			Flags		Destination Port				
RSVP_HOP Length: 12						Class Nº: 3		Class Type: 1	
Last Hop Address:						Router B			
Logical Interface Handle of the last node (LIH)									
TIME_VALUES Length: 8						Class Nº: 5		Class Type: 1	
Update Period (ms)									

Vs.: 4		iHL: 5		Service		Total Length: 60					
Identification						Flg		Fragment Offset			
Time to Live				Protocol: 46		Header Checksum					
Source Address:						Video Server					
Destination Address:						Video Client					
1		0		Type: 1		Checksum					
Send_TTL				0		Message Length: 40					
SESSION Length: 12						Class Nº: 1		Class Type: 1			
Destination Address:						Video Client					
Protocol ID				Flags		Destination Port					
RSVP_HOP Length: 12						Class Nº: 3		Class Type: 1			
Last Hop Address:						Router D					
Logical Interface Handle of the last node (LIH)											
TIME_VALUES Length: 8						Class Nº: 5		Class Type: 1			
Update Period (ms)											

RSVP RESV (Example)



Vs.: 4		IHL: 5		Service		Total Length	
Identification				Fig		Fragment Offset	
Time to Live		Protocol: 40		Header Checksum			
Source Address:				Router A			
Destination Address:				Video Server			
1	0	Type: 2		Checksum			
Send_TTL		0		Message Length			
SESSION Length: 12				Class N°: 1		Class Type: 1	
Destination Address:				Video Client			
Protocol Id		Flags		Destination protocol port			
RSVP_HOP Length: 12				Class N°: 3		Class Type: 1	
Address of the last node:				Router A			
Logical Interface Handle of the last node (LIH)							
TIME_VALUES Length: 8				Class N°: 5		Class Type: 1	
Update period (ms)							
STYLE Object Length : 8				Class N°: 8		Class Type: 1	
Flags		Style Option Vector: 0x00000A (FF)					
FLOWSPEC Length				Class N°: 9		Class Type	
FLOWSPEC object contents							
FILTER_SPEC Length: 12				Class N°: 10		Class Type: 1	
Source Address: Video Server							
Reserved		Reserved		Source protocol port			

Vs.: 4		IHL: 5		Service		Total Length	
Identification				Fig		Fragment Offset	
Time to Live		Protocol: 46		Header Checksum			
Source Address:				Router C			
Destination Address:				Router B			
1		0		Type: 2		Checksum	
Send_TTL		0		Message Length			
SESSION Length: 12				Class NP: 1		Class Type: 1	
Destination Address:				Video Client			
Protocol Id		Flags		Destination protocol port			
RSVP_HOP Length: 12				Class NP: 3		Class Type: 1	
Address of the last node:				Router C			
Logical Interface Handle of the last node (LIH)							
TIME_VALUES Length: 8				Class NP: 5		Class Type: 1	
Update period (ms)							
STYLE Object Length : 8				Class NP: 8		Class Type: 1	
Flags		Style Option Vector: 0x00000A (FF)					
FLOWSPEC Length				Class NP: 9		Class Type	
FLOWSPEC object contents							
FILTER_SPEC Length: 12				Class NP: 10		Class Type: 1	
Source Address: Video Server							
Reserved		Reserved		Source protocol port			

Vs.: 4		IHL: 5		Service		Total Length	
Identification				Fig		Fragment Offset	
Time to Live		Protocol: 46		Header Checksum			
Source Address:				Video Client			
Destination Address:				Router D			
1	0	Type: 2		Checksum			
Send_TTL		0		Message Length			
SESSION Length: 12				Class NP: 1		Class Type: 1	
Destination Address:				Video Client			
Protocol Id		Flags		Destination protocol port			
RSVP_HOP Length: 12				Class NP: 3		Class Type: 1	
Address of the last node:				Video Client			
Logical Interface Handle of the last node (LIH)							
TIME_VALUES Length: 8				Class NP: 5		Class Type: 1	
Update period (ms)							
STYLE Object Length : 8				Class NP: 8		Class Type: 1	
Flags		Style Option Vector: 0x00000A (FF)					
FLOWSPEC Length				Class NP: 9		Class Type	
FLOWSPEC object contents							
FILTER_SPEC Length: 12				Class NP: 10		Class Type: 1	
Source Address: Video Server							
Reserved		Reserved		Source protocol port			

Resource Reservation Protocol with Traffic Engineering (RSVP-TE)

- RSVP-TE is an extension of RSVP defined in:
 - RFC 3209: RSVP-TE: Extensions to RSVP for LSP Tunnels (Dec 2001)
 - RFC 5151: Resource Reservation Protocol-Traffic Engineering (RSVP-TE) Extensions (Feb 2008)
- RSVP-TE uses the resource reservation mechanism of RSVP and adds the capacity of establishing LSPs over given paths that are:
 - computed by management or
 - based on constrained-based routing provided by Extended Link State Protocols (IS-IS or OSPF with traffic engineering extensions).
 - RFC 3630: Traffic Engineering (TE) Extensions to OSPF Version 2 (Sep 2003)
 - RFC 5305: IS-IS Extensions for Traffic Engineering (Oct 2008)

Extensions to RSVP for LSP Tunnels

- The SENDER_TEMPLATE (or FilterSpec) object together with the Session object uniquely identifies an LSP tunnel (flow).
- LSP Tunnel related new objects
 - **Explicit Route:** carried in PATH and contains a sequence of variable-length items, that can be IPv4 prefixes, IPv6 prefixes or autonomous system numbers.
 - **Label Request:** carried in PATH requesting a label for a specific tunnel/flow.
 - **Label:** carried in RESV messages and contain a single label for a specific tunnel/flow.
 - **Record Route:** carried in PATH and RESV, used to collect detailed path information and useful for loop detection and diagnostics.
 - **Session Attribute:** carried in PATH, used to define the type and name of the session/tunnel/flow, also used to define priority values.
- LSP Tunnel related new object types
 - Session object new types: LSP_TUNNEL_IPv4 and LSP_TUNNEL_IPv6
 - Tspec object new types: LSP_TUNNEL_IPv4 and LSP_TUNNEL_IPv6
 - FilterSpec object new types: LSP_TUNNEL_IPv4 and LSP_TUNNEL_IPv6

Traffic Engineering with MPLS

- To fit the traffic flow throughputs in the capacity resources of the network, we need to route each traffic flow:

A – E: $A \rightarrow C \rightarrow D \rightarrow E$

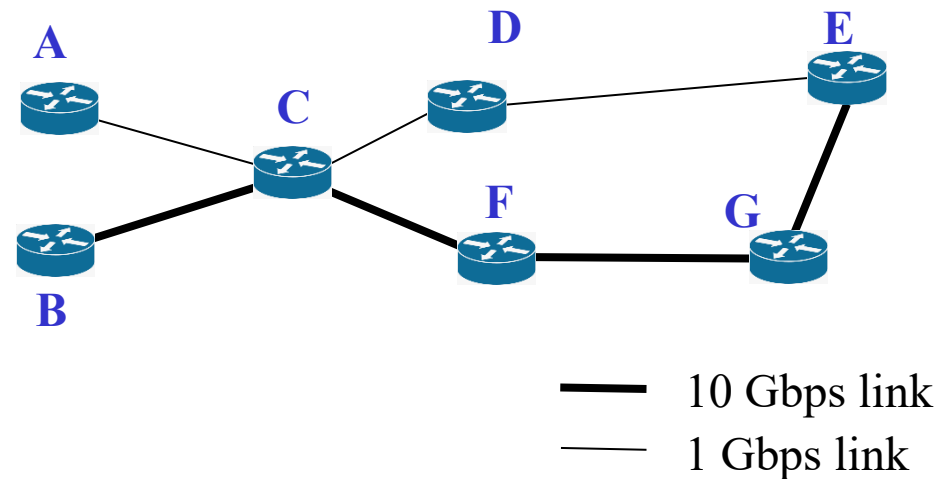
B – E: $B \rightarrow C \rightarrow F \rightarrow G \rightarrow E$

C – E: $C \rightarrow F \rightarrow G \rightarrow E$

N1	N2	N1→N2	N2→N1
A	E	1.0 Gbps	-
B	E	5.0 Gbps	-
C	E	5.0 Gbps	-

- This solution can be implemented with MPLS:

- configuring MPLS in all routers
- setting routers A, B, C and E as edge LSRs
- establishing the 3 LSPs with RSVP-TE in the above paths



Combining BGP with MPLS

- BGP and MPLS are often combined to build scalable ASs (Autonomous Systems) for Internet service providers.
 - MP-BGP acts as the control plane to advertise customer routes
 - MPLS acts as the data plane to switch packets efficiently across the core network, avoiding the need for core routers to know customer routes

- Examples:

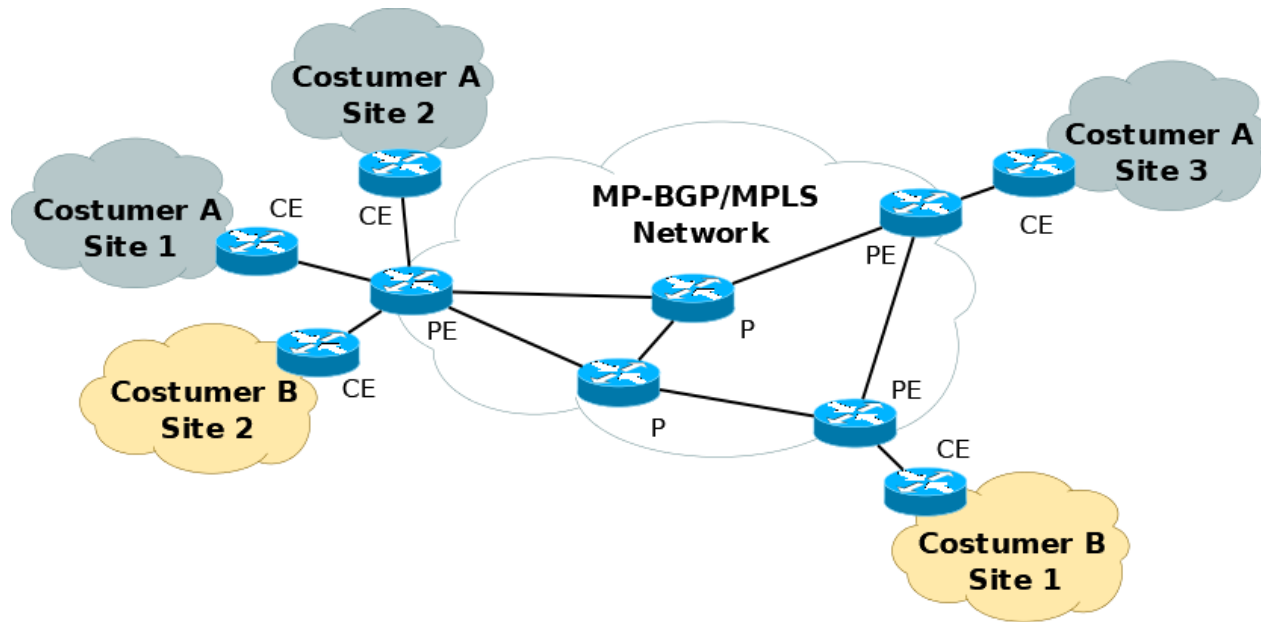
BGP 6PE (IPv6 Provider Edge over MPLS)

- It is a transition technology that allows service providers to provide IPv6 connectivity between IPv6 networks over an IPv4 MPLS backbone (i.e., it does not require dual stack backbone routers).
- Instead of IPv6-IPv4 tunnels, IPv6 is transported over MPLS LSPs.

BGP L3 VPN (BGP/MPLS Layer 3 Virtual Private Network)

- It is a technology widely used by service providers to interconnect geographically dispersed customer sites over an IP/MPLS backbone.
- It uses MP-BGP to advertise VPN routes and MPLS to forward traffic, maintaining a separate private routing table for each customer.

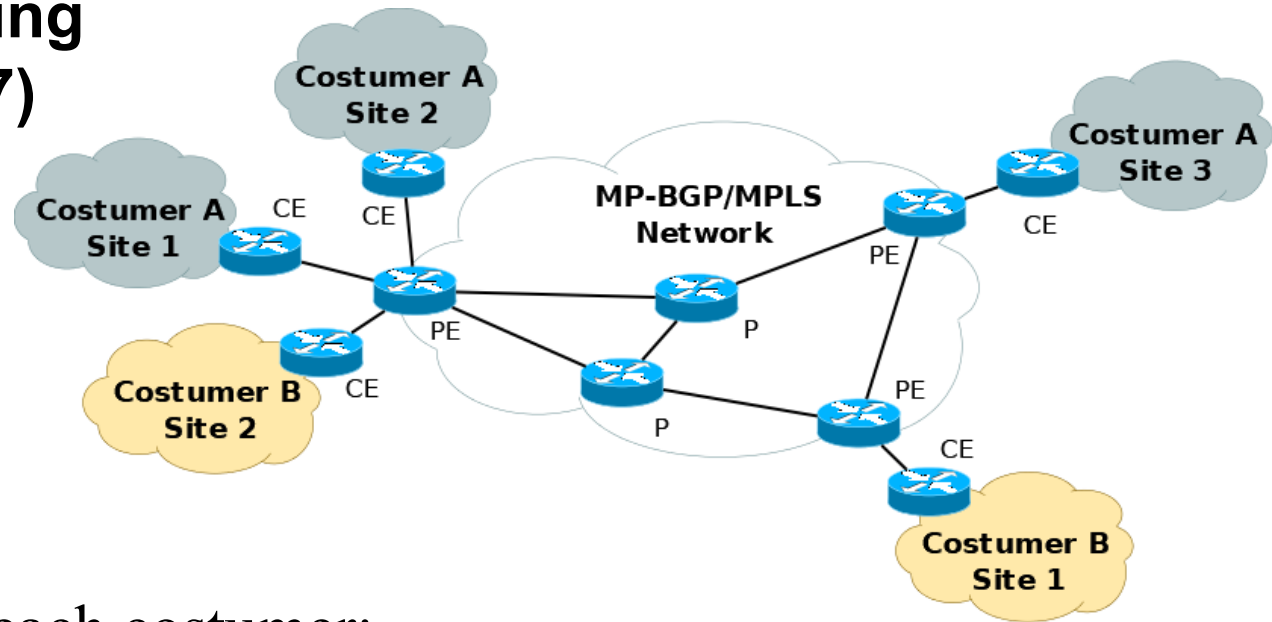
BGP L3 VPN using BGP (RFC2547)



Key Components:

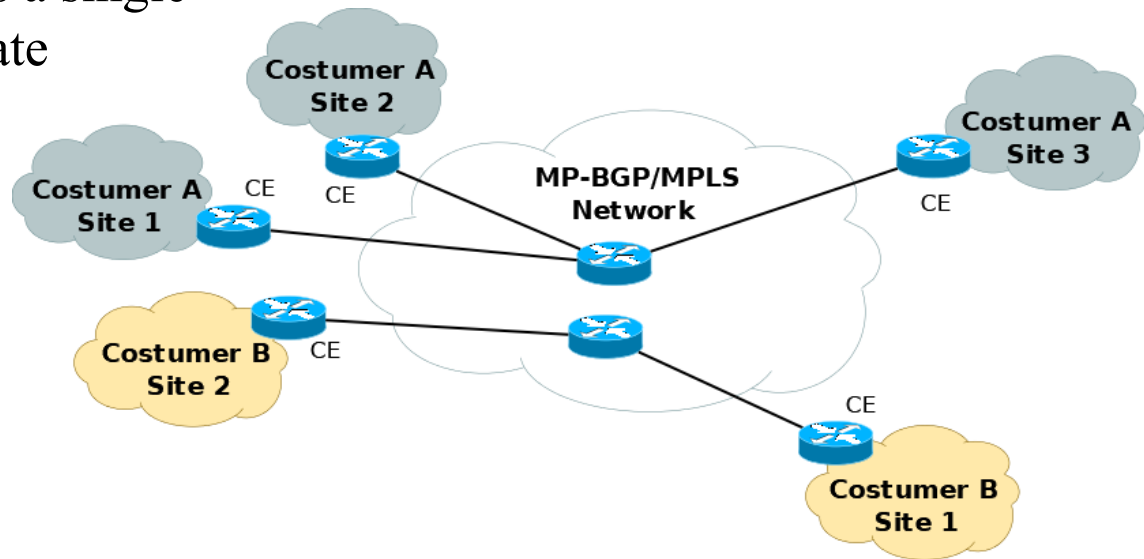
- Customer Edge (CE) Router: Connects a customer site network to a Provider Edge router, running either a dynamic routing protocol (eBGP, RIPv2, EIGRP, or OSPF) or configured with static routing
- Provider Edge (PE) Router: Responsible for VPN membership and routing, storing routes in a Virtual Routing and Forwarding (VRF) instance per customer
- Provider (P) Router: Core router that switches packets based on MPLS labels, and unaware of VPN routing

BGP L3 VPN using BGP (RFC2547)

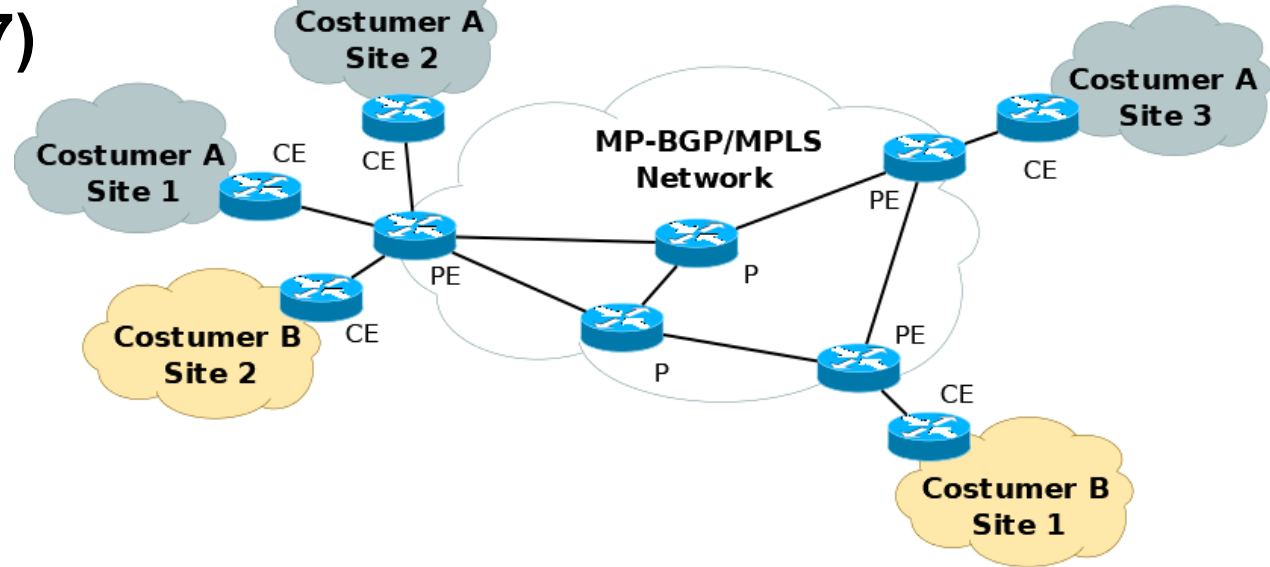


In the perspective of each customer:

- the L3 VPN service acts as a single router connecting the private networks of its sites
- customers can freely select their IP network addresses
- configuration of inter site routing does not require too much expertise



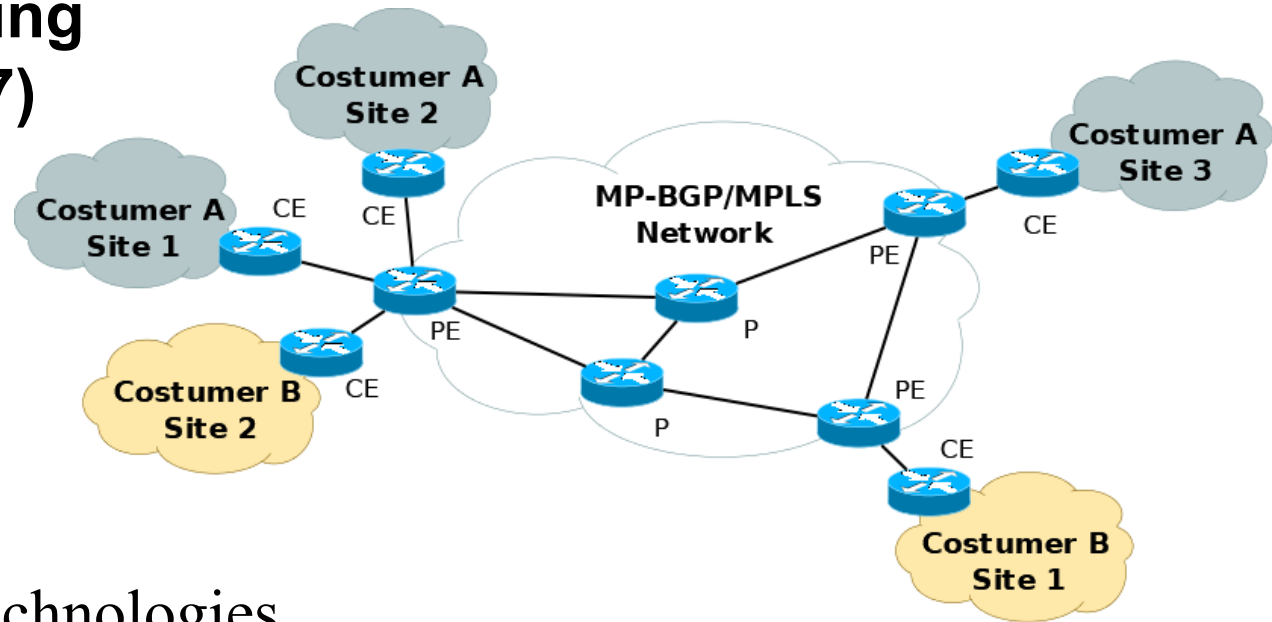
BGP L3 VPN using BGP (RFC2547)



In the perspective of the service provider, it is a scalable solution:

- Configuration of a VPN is needed only on the PE routers with connected costumer CE routers (i.e., no intervention is needed in any other router).
- An addition of a new site to a current customer with already multiple connected sites only needs configuration on the PE router to which the new CE router is connected to.
- P routers do not need to be aware of the IP networks used by each costumer.

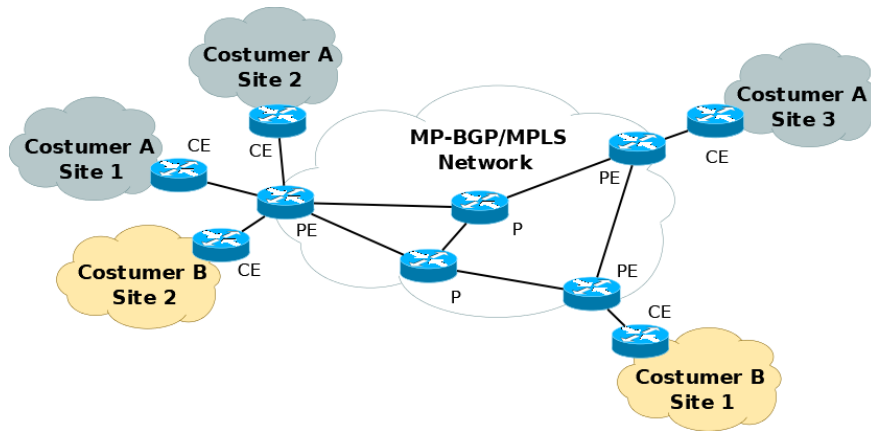
BGP L3 VPN using BGP (RFC2547)



Key Concepts and Technologies

- VRF (Virtual Routing and Forwarding): Each customer has a private routing table distributed only on the PE routers he is connected to, allowing address reuse between costumers.
- MP-BGP: Used between PE routers to distribute VPN routes of different costumers.
- RD (Route Distinguisher): An 8-byte value added to an IPv4 prefix, different for each costumer, to make prefixes globally unique in MP-BGP updates.
- RT (Route Target): An 8-byte value, different for each costumer, used to by BGP import/export routes between VRFs on PE routers.
- Labels: MPLS uses two labels, the inner VPN label to identify the destination VRF and the outer transport label to identify the appropriate LSP to traverse the core.

BGP L3 VPN using BGP (RFC2547)



```
Border Gateway Protocol - UPDATE Message
Marker: ffffffffffffffffffffffffffffffff
Length: 91
Type: UPDATE Message (2)
Withdrawn Routes Length: 0
Total Path Attribute Length: 68
Path attributes
  Path Attribute - ORIGIN: INCOMPLETE
  Path Attribute - AS_PATH: empty
  Path Attribute - MULTI_EXIT_DISC: 0
  Path Attribute - LOCAL_PREF: 100
```

1.

```
  Path Attribute - EXTENDED_COMMUNITIES
    Flags: 0xc0: Optional, Transitive, Complete
    Type Code: EXTENDED_COMMUNITIES (16)
    Length: 8
    Carried extended communities: (1 community)
      Community Transitive Two-Octet AS Route Target: 200:1
```

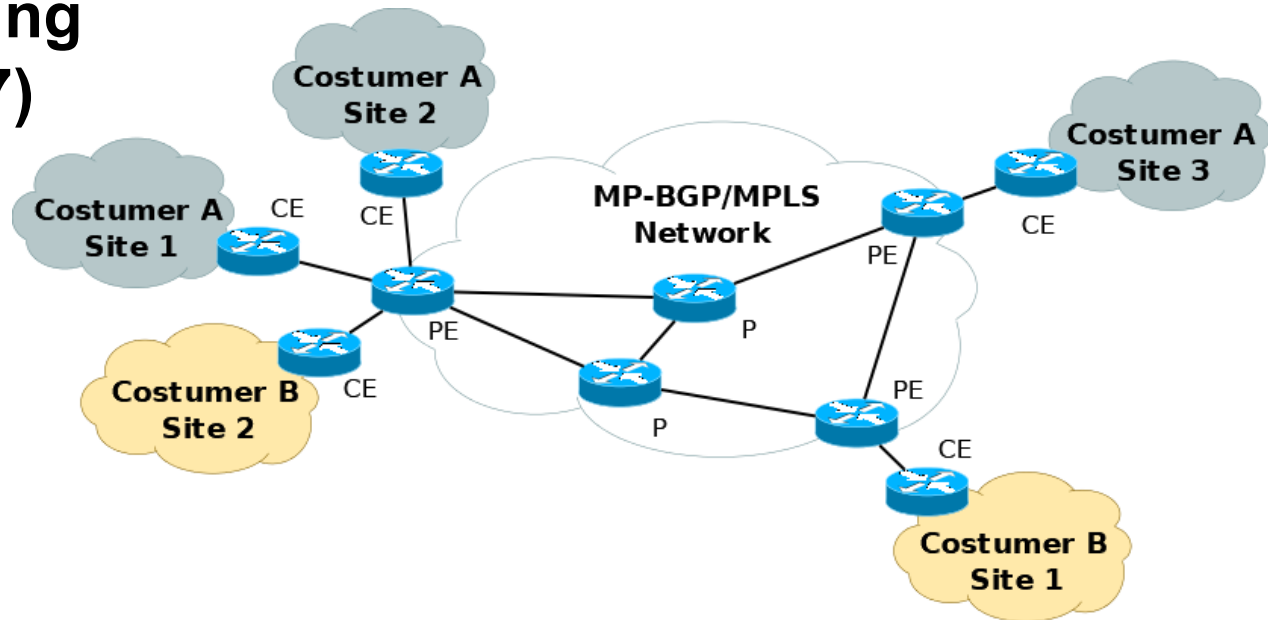
2.

```
  Path Attribute - MP_REACH_NLRI
    Flags: 0x80: Optional, Non-transitive, Complete
    Type Code: MP_REACH_NLRI (14)
    Length: 33
    Address family: IPv4 (1)
    Subsequent address family identifier: Labeled VPN Unicast (128)
    Next hop network address (12 bytes)
    Subnetwork points of attachment: 0
    Network layer reachability information (16 bytes)
      Label Stack=24 (bottom) RD=200:1, IPv4=192.1.1.0/25
```

Each PE router sends MP-BGP update messages to all other PE routers announcing the VPN IPv4 prefixes of its directly connected customer sites.

1. RT (Route Target) value is sent in the EXTENDED_COMMUNITIES attribute, it identifies the customer VPN of the announced IPv4 prefix
2. MP_REACH_NLRI attribute includes:
 - the announced VPN IPv4 prefix
 - the RD (Route Distinguisher) value, added to allow the same IPv4 prefix to be used by different customers
 - the inner VPN label, used to identify the customer VPN at IP packet reception

BGP L3 VPN using BGP (RFC2547)



IP Packet Forwarding between sites of a costumer:

1. **Ingress:** CE sends an IP packet to PE. PE classifies the packet into a VRF based on the incoming port, adds the inner VPN label and the outer MPLS label.
2. **Core:** P routers switch the packet using the outer label towards the egress PE.
3. **Egress:** The egress PE removes the outer label, uses the inner label to identify the outgoing interface/VRF, removes the inner label, and sends it to the appropriate CE of the customer.

In the last step, if PHP (Penultimate Hop Popping) is set, the last P router removes the inner label, and the packet reaches the egress PE router with a single label identifying the outgoing interface/VRF.